Credit-Aware Counterfactual Rollout Allocation for LLM Reasoning

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Abstract

Reinforcement learning (RL) for large language model (LLM) reasoning commonly provides feedback only at the end of a reasoning trajectory, making it difficult to determine which intermediate decisions caused the final outcome. Tree-based approaches address this problem by sharing intermediate reasoning states across multiple trajectories, but when the number of rollouts is fixed, an important question remains: where should additional trajectories be generated to obtain useful counterfactual evidence? We propose a two-stage adaptive rollout framework that separates exploration from exploitation. In the first stage, $n - k$ trajectories, where n is the total rollout budget, are used to construct an exploratory reasoning tree using surprisal, following the uncertainty-driven expansion principle of TreeRL. In the second stage, the initial tree is analyzed using a perplexity-based novelty proxy and a tree-based credit signal. Candidate states are first filtered according to their potential to produce continuations different from those already observed. The remaining k rollouts are then allocated among these candidates according to their estimated consequentiality, with additional trajectories generated from the corresponding parent states to obtain targeted counterfactual evidence. The completed tree is finally used to derive variable, node-level credit estimates for policy optimization. We hypothesize that separating novelty-based candidate selection from credit-based allocation can provide more informative counterfactual evidence without increasing the total rollout budget. Our code is open-sourced at **TODO: add the link of whatever we finalize as our github repo AND MAKE THE REPO PUBLIC**.

1 Introduction

Reinforcement learning (RL) has become an effective approach for improving the reasoning capabilities of large language models (LLMs). However, reasoning tasks often provide feedback only through a terminal reward, such as whether the final answer is correct. This creates a fundamental *credit assignment* problem: a long reasoning trajectory may contain many intermediate decisions, while the final reward does not indicate which decisions were responsible for the outcome.

A natural way to obtain more localized credit is to generate multiple trajectories that share intermediate reasoning states. When trajectories diverge from the same state and receive different terminal rewards, their outcomes provide counterfactual evidence about the consequences of decisions at that state. Recent methods exploit this structure for credit assignment and exploration, including TEMPO (Tran et al., 2025), TreePO (Li et al., 2025), and TreeRL (Hou et al., 2025). Other works investigate adaptive rollout allocation (Zhang et al., 2026; Zou et al., 2026) and the use of counterfactual reasoning paths for reducing credit-assignment variance (Ding et al., 2026).

However, generating trajectories can be computationally expensive, making the rollout budget a limited resource. This motivates the following question:

> *Given a fixed rollout budget, where should additional trajectories be generated to obtain the most useful counterfactual evidence for credit assignment?*

Generating all trajectories according to a single exploration criterion may spend computation on states that are uncertain but do not benefit substantially from additional sampling. Conversely, immediately allocating rollouts according to observed rewards may overexploit states before sufficient alternative outcomes have been explored.

We therefore propose a two-stage strategy that explicitly separates *exploration* from *exploitation*. The first stage uses $n - k$ trajectories to construct an exploratory reasoning tree using surprisal, following the uncertainty-driven expansion principle

of TreeRL (Hou et al., 2025). The second stage uses the resulting tree to identify states where additional sampling may produce novel continuations and then allocates the remaining k trajectories according to observed consequentiality. Rather than combining novelty and credit into a single numerical score, novelty is first used to filter the candidate states, after which credit determines how the reserved budget is distributed among the remaining candidates.

2 Background

2.1 Tree-Based Reasoning and Credit Assignment

Tree-based rollout methods represent multiple reasoning trajectories as a shared tree. Each internal node corresponds to an intermediate reasoning state, while different children represent alternative continuations from that state. This structure makes it possible to compare trajectories that share a common reasoning prefix.

Consider a reasoning trajectory

$$\tau = (v_1, \ldots, v_m), \tag{1}$$

where each v_i represents a reasoning segment and the trajectory receives a terminal reward $R(\tau)$. For a node v, let $\mathcal{L}(v)$ denote its descendant leaf trajectories. A simple empirical value estimate is

$$V(v) = \frac{1}{|\mathcal{L}(v)|} \sum_{\ell \in \mathcal{L}(v)} R(l). \tag{2}$$

When multiple trajectories share v, this value summarizes the observed outcomes associated with that reasoning state. The same tree structure can therefore be used both to obtain localized credit signals and to determine where additional rollout computation should be spent.

3 Related Work

Tree-based reasoning and credit assignment. TEMPO (Tran et al., 2025) exploits shared tree structure to estimate intermediate values from descendant outcomes and obtain more localized credit signals. TreePO (Li et al., 2025) uses segment-level tree rollouts for reasoning optimization, while TreeRL (Hou et al., 2025) performs on-policy tree search by expanding uncertain intermediate states. Our method similarly uses a shared reasoning tree, but explicitly divides the rollout budget between an exploratory stage and a subsequent targeted expansion stage.

Adaptive rollout allocation. IGRPO (Zhang et al., 2026) investigates information-gain-based adaptive rollout allocation in a tree-structured rollout framework, while TRACE (Zou et al., 2026) studies adaptive allocation of a fixed rollout budget. These works motivate the idea that rollout computation should be allocated selectively rather than uniformly. Our approach similarly treats rollouts as a limited resource, but uses a two-stage procedure in which the first stage builds an exploratory tree and the second stage uses novelty and observed consequentiality to allocate additional trajectories.

Counterfactual reasoning paths. The work *Reducing Credit-Assignment Variance Through Counterfactual Reasoning Paths* (Ding et al., 2026) investigates how alternative reasoning paths can provide useful information for reducing variance in credit assignment. Our approach addresses the complementary question of where such additional counterfactual paths should be generated when only a fixed number of rollouts is available.

4 Proposed Method

4.1 Problem Formulation

For each problem x, let n denote the total rollout budget. We divide it into two portions:

$$n = (n - k) + k. \tag{3}$$

The first $n - k$ trajectories are used for *exploration*, while the remaining k trajectories are reserved for *credit-guided counterfactual expansion*.

Each tree node represents a reasoning segment rather than an individual token. Thus, a trajectory consisting of segments

$$v_1, v_2, \ldots, v_m$$

forms a path in the tree, and trajectories sharing a prefix reuse the corresponding nodes.

4.2 Phase I: Entropy-Guided Exploration

The first phase constructs an initial tree T_0 using $n - k$ trajectories. The goal of this phase is exploration rather than exploitation. Terminal rewards are therefore not used to determine where trajectories are generated.

Instead, we follow the uncertainty-driven expansion principle of TreeRL (Hou et al., 2025), using local token-level surprisal to identify reasoning states with high uncertainty.

4.2.1 Initial Trajectory

We first generate a complete reasoning trajectory from the original problem x. Every reasoning segment becomes a node in T_0, so the initial tree consists of a single path from the root to a terminal leaf.

For subsequent rollouts, eligible reasoning states already present in the tree are considered as possible branching locations.

4.2.2 Surprisal

Let c_v denote the model context at reasoning state v, and let a be the selected next token. The model assigns a probability to this token as

$$p_\theta(a \mid c_v). \tag{4}$$

We compute the surprisal at v as

$$H(v) = -\log p_\theta(a \mid c_v). \tag{5}$$

High surprisal indicates that the model assigns a low probability to the specific next token. Such states are therefore treated as promising locations for exploration, since the model exhibits greater surprise or uncertainty about this immediate continuation.

At each rollout step, we select an eligible state with high surprisal and generate a complete continuation from that state. The new trajectory shares the existing prefix with the tree and creates a new branch at the selected reasoning state.

After adding the trajectory, the tree is updated and the entropy values of eligible states are recomputed as needed. This process continues until exactly $n - k$ trajectories have been generated.

No terminal reward is used during this phase, keeping the initial exploration independent of observed task success.

4.3 Phase II: Novelty-Aware Credit-Guided Expansion

After the exploratory tree has been constructed, terminal rewards become available. The remaining k rollouts are reserved for targeted counterfactual expansion.

The goal of Phase II is to identify reasoning states where another rollout is both potentially informative and consequential. We therefore separate the two considerations rather than combining them into a single score.

First, a perplexity-based measure is used to filter candidate states according to their potential to yield novel continuations. Second, the remaining candidates are ranked through a tree-based credit signal, which determines where the reserved rollout budget should be allocated.

4.3.1 Perplexity-Based Novelty

Suppose node v has already produced a set of observed continuations $\mathcal{L}(v)$. For each continuation ℓ, let $S_{\ell,v}$ denote the suffix following v.

If this suffix contains $L_{\ell,v}$ tokens, its average negative log-likelihood is

$$H_{\ell,v} = -\frac{1}{L_{\ell,v}} \sum_{t=1}^{L_{\ell,v}} \log p_\theta(x_t \mid c_v, x_{<t}). \tag{6}$$

The corresponding continuation perplexity is

$$PPL_{\ell,v} = \exp(H_{\ell,v}). \tag{7}$$

We use continuation perplexity as a proxy for the potential to obtain novel continuations from v. The intuition is that if the continuations already observed from a state are not strongly concentrated under the model, additional samples may have greater potential to differ from those already observed.

Importantly, perplexity does not directly measure the probability that a future sample will be unique. It is therefore treated only as a proxy for novelty.

For each node with multiple observed continuations, we compute the mean continuation perplexity:

$$D(v) = \frac{1}{|\mathcal{L}(v)|} \sum_{\ell \in \mathcal{L}(v)} PPL_{\ell,v}. \tag{8}$$

Higher $D(v)$ indicates greater perplexity among the observed continuations and is therefore interpreted as greater estimated novelty potential.

The mean perplexity formulation will serve as our initial novelty measure. Alternative statistics, such as the variance of continuation perplexity, can be investigated in ablation experiments.

4.3.2 Novelty-Based Candidate Filtering

Let $\mathcal{V}(T_0)$ denote the set of eligible non-root reasoning states in the initial tree. We retain the

top fraction ρ of states according to their novelty scores:

$$\mathcal{V}_{\text{novel}} = \text{Top}_{\rho}\left(\{D(v) : v \in \mathcal{V}(T_0)\}\right), \quad (9)$$

where $\rho \in (0, 1]$ determines the fraction of candidate nodes retained.

Using a relative top-ρ criterion prevents the candidate set from becoming empty when all states have relatively low novelty scores. Thus, even when the absolute novelty of all states is limited, the states with the greatest relative potential for additional variation remain eligible for credit-based allocation.

This gives the novelty component a clear role:

> Which states are worth considering for another rollout because they may still produce different behavior?

The credit signal then determines which of these candidates should receive the limited additional budget.

4.3.3 Tree-Based Value Estimation

For each node v, we estimate its empirical value from the terminal rewards of its descendant trajectories:

$$V(v) = \frac{1}{|\mathcal{L}(v)|} \sum_{\ell \in \mathcal{L}(v)} R(l). \quad (10)$$

For binary mathematical reasoning rewards, $V(v)$ corresponds to the empirical success rate of trajectories passing through v.

4.3.4 Local Consequentiality

For each candidate node v, we measure the change in estimated value across the reasoning decision leading to that node:

$$\delta(v) = V(v) - V(\text{parent}(v)). \quad (11)$$

The parent state represents the reasoning state at which the decision leading to v was made. Thus, $\delta(v)$ provides a local signal for how the observed outcome value changes across that decision.

We use the magnitude of this difference as the consequentiality score:

$$C(v) = |\delta(v)|. \quad (12)$$

A larger value indicates that the observed outcomes differ more substantially across the corresponding decision boundary.

4.3.5 Credit-Guided Allocation

After novelty filtering, only nodes in $\mathcal{V}_{\text{novel}}$ are considered for the remaining rollouts. Their consequentiality scores are converted into an allocation distribution:

$$p(v_i) = \frac{\exp(C(v_i)/\tau)}{\sum_{v_j \in \mathcal{V}_{\text{novel}}} \exp(C(v_j)/\tau)}, \quad (13)$$

where τ is a temperature parameter controlling the concentration of the allocation distribution.

For each of the remaining k rollouts, a node v is sampled according to this distribution. We then expand the *parent* of the selected node rather than v itself. The parent represents the reasoning state at which the original decision was made. Generating an additional continuation from this state therefore provides further evidence about the consequences of that decision.

For example, if the initial tree contains

$$p(v) \to v \to \cdots \to \ell_1,$$

the expanded tree may contain

$$p(v) \to \begin{cases} v \to \cdots \to \ell_1, \\ v' \to \cdots \to \ell_2. \end{cases}$$

The original trajectory is preserved, while the new trajectory provides additional counterfactual evidence from the same decision state.

After each newly generated trajectory is added, the relevant tree statistics can be recomputed so that subsequent allocation decisions incorporate the newly observed evidence. This process continues until all k reserved rollouts are consumed.

4.4 Final Credit Assignment and Policy Optimization

After counterfactual expansion, the tree values are recomputed to incorporate all n trajectories. The complete tree is then used to obtain variable, node-level credit estimates.

Rather than assigning a uniform trajectory-level advantage to every reasoning decision, we derive credit signals from the tree structure and local value differences. These signals are incorporated into the advantage estimates used for policy optimization.

Thus, the final training procedure is a policy optimization method with *tree-derived, variable-credit advantage estimates*. The credit assigned to an intermediate reasoning state can therefore

depend on its observed position and consequences within the completed tree.

4.5 Algorithm

Algorithm 1 summarizes the complete rollout allocation procedure.

The procedure uses exactly $n - k$ exploratory trajectories and k targeted trajectories, maintaining the fixed total rollout budget n.

5 Dataset and Experimental Setup

5.1 Datasets

We will initially evaluate the method on verifiable mathematical reasoning tasks. Mathematical reasoning provides deterministic terminal rewards based on final-answer correctness, allowing controlled evaluation of rollout allocation and credit assignment.

TODO: replace with what we actually used in the final code for which we finally decide to show the results

5.2 Model and Training

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6 Results

TODO: Add whatever final results we decide to show.

7 Future Evaluation Plan

We will evaluate the method along three dimensions: reasoning performance, counterfactual allocation quality, and rollout efficiency.

Reasoning performance. We will measure final-answer accuracy on held-out mathematical reasoning benchmarks and compare against relevant tree-based approaches such as TEMPO, TreePO, and TreeRL under matched rollout budgets.

Counterfactual allocation quality. We will evaluate whether states selected for additional expansion provide more informative outcome variation than states selected by alternative strategies. On a smaller subset, we will generate substantially more rollouts from candidate states to approximate their possible outcome distributions. This provides an approximate oracle for evaluating whether the selected states exhibit greater downstream outcome variation.

Phase-I exploration ablations. We will compare local-entropy-based exploration against random branching and other uncertainty-based alternatives where feasible.

Phase-II allocation ablations. We will compare:

- credit-guided allocation without novelty filtering;
- novelty filtering followed by credit-guided allocation;
- novelty-only allocation;
- random or uniform allocation.

These comparisons will directly test whether the novelty filter provides benefits beyond the credit signal itself.

Novelty-score ablations. We will compare mean continuation perplexity against alternative statistics such as the variance of continuation perplexity.

Candidate fraction. We will vary the fraction ρ of nodes retained by the novelty filter to study the effect of candidate-set size.

Budget efficiency. We will evaluate several total rollout budgets n and reserved-budget ratios k/n, measuring whether the proposed method provides better reasoning performance or credit localization at the same total computational budget.

8 Expected Contributions

The project aims to make three contributions.

First, we formulate adaptive rollout allocation as a *credit-aware exploration problem*, treating additional trajectories as a limited resource for obtaining counterfactual evidence.

Second, we propose a two-stage rollout allocation framework that separates uncertainty-driven exploration from novelty-aware, credit-guided exploitation. The first stage constructs an exploratory tree using surprisal, while the second stage filters candidate states according to a perplexity-based novelty proxy and allocates the reserved rollout budget according to observed consequentiality.

Third, we investigate whether targeted counterfactual expansion can improve variable, node-level credit assignment without increasing the total rollout budget.

9 Limitations and Future Work

The Phase-I exploration strategy follows the uncertainty-driven expansion principle of TreeRL and therefore relies on surprisal as a proxy for exploratory potential. Surprisal reflects uncertainty over immediate model predictions, but does not directly measure the downstream usefulness of the resulting continuation for credit assignment.

The Phase-II novelty signal is based on continuation perplexity and is therefore only a proxy for the probability of obtaining a previously unobserved continuation. High perplexity of observed continuations does not guarantee that a new rollout will be qualitatively different. Future experiments will investigate alternative measures of novelty and diversity.

The consequentiality signal

$$\delta(v) = V(v) - V(\text{parent}(v)) \tag{14}$$

is also only one possible measure of decision importance. Future experiments may investigate alternative signals based on value uncertainty, child-value disagreement, or estimates of the value of an additional rollout.

Finally, the initial evaluation focuses on mathematical reasoning because deterministic rewards enable controlled experimentation. Extending the framework to tasks with richer process rewards or interactive environments is left for future work.

10 Tentative Timeline

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- **Week 1, Oct 2026:** Implement segment-level tree construction and entropy-guided exploration.
- **Week 2, October 2026:** Implement novelty filtering and credit-guided counterfactual allocation.
- **Week 3, October 2026:** Run baseline and ablation experiments.
- **Week 4, October 2026:** Analyze rollout allocation, credit localization, and reasoning performance.

Algorithm 1 Novelty-Aware Credit-Guided Rollout Allocation

Require: Input problem x, rollout budget n, reserved budget k, temperature τ, candidate fraction ρ

Initialize tree $T_0 \leftarrow \emptyset$
Generate a complete trajectory ℓ_1 from x
Add ℓ_1 to T_0
for tree $i = 1, \ldots, M$ **do**
  for iteration $l = 1, \ldots, L$ **do**
    Compute surprisal $H(v)$ for eligible nodes
    Select top-N high-surprisal branching nodes
    for each selected branching node v **do**
      Generate T complete continuations from v
      Add the trajectories to T_0
    end for
    Update the tree
  end for
end for
for each eligible node $v \in \mathcal{V}(T_0)$ **do**
  Compute continuation perplexity statistics
  Compute novelty score $D(v)$
end for
Construct $\mathcal{V}_{\text{novel}}$ from the top-ρ nodes according to $D(v)$
for each $v \in \mathcal{V}_{\text{novel}}$ **do**
  Compute $V(v)$
  Compute $\delta(v) = V(v) - V(\text{parent}(v))$
  Compute $C(v) = |\delta(v)|$
end for
Construct

$$p(v_i) = \frac{\exp(C(v_i)/\tau)}{\sum_{v_j \in \mathcal{V}_{\text{novel}}} \exp(C(v_j)/\tau)}$$

for $j = 1, \ldots, k$ **do**
  Sample $v \sim p(v)$
  Set expansion state to $\text{parent}(v)$
  Generate an independent continuation from the expansion state
  Add the trajectory to T_0
  Update relevant tree statistics
end for
Recompute tree values and node-level credit signals
Construct variable-credit advantage estimates
Perform policy optimization
return Completed tree and updated policy

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